

IMAGINARY ELECTRIC COMPANY

Power Generation Division - Plant Integration

Alpilla 600 MW Combined Cycle Power Plant - 1 x IE-9H.02 multi-shaft power island

Seawater-Cooled Surface Condenser - Datasheet

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	Prepared	Checked	Approved
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CASE STUDY - Imaginary Electric, its products and all data are fictional.

1. Thermal design (HEI 11th edition)

Item	Value
Type	single-shell, two-pass surface condenser, divided water boxes (two halves), side-mounted to the ST
Design duty at SRC-NG-100	289.96 MW
Condenser pressure at SRC	34.44 mbar(a) (guarantee 35.5 mbar(a); ER-07.08 limit 36 mbar(a))
Seawater inlet / outlet	16.0 / 23.0 degC
Circulating water flow	36,050 m3/h (10,179 kg/s)
Seawater properties	density 1,016.5 kg/m3, cp 4.07 kJ/kg K (salinity 22 PSU, ALP-OWN-MAR-001)
Surface area (incl. 5 % plugging margin)	15,200 m2
Tubes	titanium Grade 2 welded, OD 25.4 mm, 0.5 mm wall (0.7 mm in the impingement zone), effective length 9.35 m, 20,400 tubes
Tube velocity	2.1 m/s (ER-07.07: 1.8 - 2.2 m/s)
Cleanliness factor	0.85
Seawater pressure loss	0.62 bar (water boxes and tubes, clean)
Tube sheets / water boxes	titanium-clad carbon steel / rubber-lined carbon steel with impressed-current cathodic protection
Hotwell	30 m3 working volume (3.9 min at SRC condensate flow)
Dissolved oxygen at hotwell outlet	<= 20 ug/l above 50 % load
Bypass operation	100 % HP/IP/LP bypass at SRC: 365 MW; maximum (cold ambient, LDO): 395 MW; seawater rise then 8.8 - 9.5 K (transient)
Vacuum system	2 x 100 % liquid-ring vacuum pumps 75 kW (holding) + hogging by both pumps in parallel (30 min to 150 mbar)
Tube cleaning	tube cleaning ball system and debris filters: not included (by EPC with the CW system); condenser inlet water boxes prepared for ball injection and strainer connection

2. Performance at other seawater temperatures (base load, SRC steam flow)

Seawater inlet degC	6.9	11.0	16.0	22.0	24.0	27.0
Condenser pressure mbar(a)	19.8	25.5	34.4	48.7	54.4	64.2

At 27 degC seawater the condenser pressure is 64 mbar(a), well inside the ST continuous limit of 120 mbar(a) (ER-07.08).

3. Weights

Item	Weight (t)
Condenser shell with tube bundles (shipped in 4 modules)	4 x 95
Operating weight (water-filled)	1,650
Flooded weight (hydrotest)	2,300